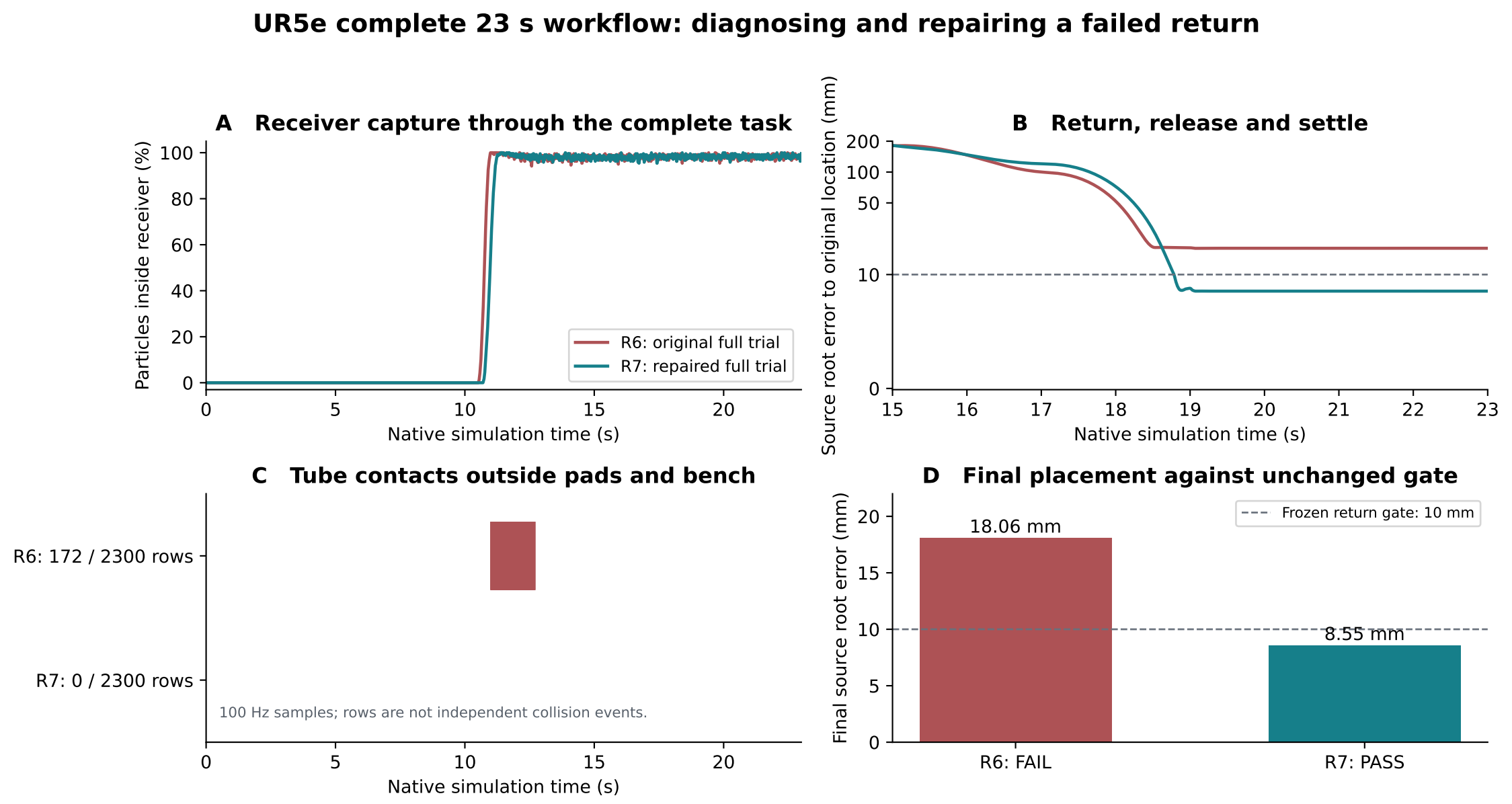
UR5e complete 23 s workflow: diagnosing and repairing a failed return
A Receiver capture through the complete task
Particles inside receiver (%)
Native simulation time (s)
R6: original full trial
R7: repaired full trial
B Return, release and settle
Source root error to original location (mm)
Native simulation time (s)
C Tube contacts outside pads and bench
R6: 172 / 2300 rows
R7: 0 / 2300 rows
100 Hz samples; rows are not independent collision events.
Native simulation time (s)
D Final placement against unchanged gate
Final source root error (mm)
Frozen return gate: 10 mm
18.06 mm
8.55 mm
R6: FAIL
R7: PASS
One trial per version. R7 changes tilt limit and native-pose return feedback and uses CUDA; R6 uses CPU.
This comparison does not isolate a mechanism or establish robustness, perception or learned control.